ARTCRAFT FLEET TECHNICAL BROCHURE · SERIES #20

BuchCraft v0.2.1

Austrian double-entry bookkeeping as a local-first craft app. Pure Rust,



v0.2.1	4 Crates	5153	6-Gate
RELEASE VERSION	WORKSPACE TIERS	PURE-RUST LOC	ARTCRAFT VERIFIED

Repository: `D:\Dev\repos\buchcraft` · **Standard:** `CRAFT_BROCHURE_STANDARD.md` · **Compiled via:** `BookCraft v0.5.0`

Austrian double-entry bookkeeping as a local-first craft app. Pure Rust, SQLite (WAL), one binary that is both a CLI and an MCP server. Part of the back-office craft suite (plan: `mcp-central-docs/projects/craft-fleet-expansion/BACKOFFICE_CRAFT_SUITE.md`).

Replaces the bookkeeping part of BMD NTCS, RZL, DATEV or sevDesk for a small Austrian business. Does **not** file anything with FinanzOnline, is not a Registrierkasse (RKSV) and does no payroll.

- How to use it day to day: HELP.md
- What it is for and where it stops: docs/PRD.md
- Every MCP tool: docs/TOOLS.md (generated, checked by a test)
- Changes: CHANGELOG.md

What is real in v0.2.1

Feature	State
EKR chart (seed + own accounts)	Real. Seed numbers are a starting point; check them against your Steuerberater's chart.
Balanced entries, integer cents	Real. Sum of lines must be exactly zero.
VAT keys U20/U10/U13/V20/V10/V13/IGE20	Real. You book NET lines with a key; the engine adds the tax line. You cannot book VAT twice.
Immutability + Storno	Real. SQLite triggers block UPDATE/DELETE on booked rows; corrections are linked Storno entries.
Month close	Real. Bookings dated in a closed month are refused.
Saldenliste, Kontoblatt, journal	Real.
U30 pre-fill (KZ 000/006/022/029/060/065/070/072/095)	Real computation from booked entries. Not a filing.
Bank CSV import	Real parser with German/English header aliases, dedup on re-import. **Not yet verified against a real George export.**
Drafts from other craft apps (bus)	Real for `asset.depreciation_due`, verified end to end with assetcraft (`afa --emit` -> draft -> approve -> booked). Drafts book only on approval.
Demo company	Labelled demo (`demo: true` everywhere), only on an empty ledger.
Desktop app (egui)	Real: overview with KPIs and 6-month chart, bank inbox (pick account + tax key, book), gross booking form with live net/tax preview, journal with Storno, Saldenliste, Kontoblatt, U30 with month close, drafts. De/En, light/dark. No egui_kittest golden snapshots yet.

Desktop app

```
.\start.bat                       # builds and opens the window
buchcraft-app --db .\demo.db --lang de --view bank
buchcraft-app --smoke             # headless status check, no window
buchcraft-app --view u30 --screenshot u30.png   # render one PNG and exit
```

!Overview

!Bank inbox

!U30

Quick start

```
cargo build --release
$b = ".\target\release\buchcraft.exe"   # or D:\Dev\cargo-target\release
& $b --db .\demo.db demo-seed
& $b --db .\demo.db u30 2026-10
& $b --db .\demo.db balance --from 01.10.2026 --to 31.10.2026
& $b --db .\demo.db bank unmatched
& $b --db .\demo.db accounts add 7300 "Reisekosten" --kind expense --tax-key V10
& $b --db .\demo.db bank book 3 --account 7300 --tax-key V10
```

Add `--json` to any verb for machine output. Default database:

`$env:BUCHCRAFT_DB` or `%LOCALAPPDATA%\buchcraft\buchcraft.db`.

Who uses it, for what

User	Prompt to their agent	Tools that run
Sole trader (e.U.)	"Import my George export and suggest accounts for the unbooked lines"	`buch_bank_import`, `buch_bank_unmatched`, `buch_bank_book`
Small GmbH office	"What is our VAT payable for October?"	`buch_u30`
Bookkeeper	"Reverse invoice AR-2026-0002, wrong amount, then rebook it"	`buch_entry_reverse`, `buch_entry_book`
Steuerberater hand-off	"Export the October journal as CSV for the tax adviser"	`buch_export_csv`
assetcraft user	"Book this year's depreciation"	`buch_inbox_drain`, `buch_drafts`, `buch_draft_approve`

MCP

```
{
  "mcpServers": {
    "buchcraft": {
      "command": "D:\\Dev\\cargo-target\\release\\buchcraft.exe",
      "args": ["mcp", "--allow-path", "D:\\Books"]
    }
  }
}
```

File tools (`buch_bank_import`, `buch_export_csv`) only work inside `--allow-path` directories in MCP mode. 22 tools, listed in docs/TOOLS.md (`buchcraft tools --markdown`). Amounts in tool arguments are DE strings (`"1.200,00"`) or integer `*_cents`; floats are refused.

Gates

`just ci` = clippy `-D warnings` (unwrap/expect/panic denied), tests, fmt.

v0.2.1: 28 tests (including the docs/TOOLS.md sync test), live MCP stdio probe verified (stdout pure JSON-RPC). `just docs` regenerates docs/TOOLS.md.

Product Requirements & Architecture Specification

What it is

Austrian double-entry bookkeeping (Einnahmen-Ausgaben is not the model; this is doppelte Buchhaltung on the EKR chart) for a small business, local-first. Pure Rust, ArtCraft pattern: one engine, CLI + MCP + egui window. The hub of the back-office craft suite (`BACKOFFICE_CRAFT_SUITE.md`, section 4.1).

Users

- The owner or bookkeeper doing monthly books and the UVA pre-fill.
- Agents over MCP: propose accounts and tax keys for unmatched bank lines, prepare drafts; a human approves before anything is booked.
- The Steuerberater receives the Saldenliste and journal export.

Jobs

1. Book correctly and immutably, with VAT computed by the engine.
2. Get bank lines in and matched without retyping.
3. Produce U30 figures, Saldenliste and Kontoblatt at month end.
4. Accept bookings proposed by other craft apps (AfA from assetcraft) as drafts.

Today (v0.2.0)

- Ledger on SQLite WAL with immutability triggers; entries balance to the cent; the engine writes all tax lines; Storno with two-way links; month close.
- EKR seed chart with tax keys (V20/V10/V13/U20/U10/U13/IG) and U30 Kennzahlen.
- Bank CSV import (Erste George layouts), dedup, booking of bank lines.
- Drafts inbox fed by the craft-biz bus (`asset.depreciation_due`), verified end to end with assetcraft.
- 22 MCP tools (TOOLS.md, generated and test-checked), CLI, egui app with 8 views (De/En, light/dark, screenshot mode).

Boundaries

- No FinanzOnline submission, no RKSV / Registrierkasse, no payroll, no UGB balance sheet layout (Saldenliste only), EUR only.
- Agents never book directly from external input: drafts need human approval.
- Not a replacement for the Steuerberater's review.

Gaps (honest)

- Bank CSV aliases not verified against a real George export.
- EKR seed account numbers need a Steuerberater check.
- CAMT.053 import not built.
- No golden-snapshot GUI tests, no democraft recipes, no installer yet.
- Synthetic data only.

Next

1. Verify the bank importer with one real George export.
2. egui_kittest snapshots for the 8 views; right-align amount columns.
3. Steuerberater review of the seed chart and U30 mapping.

Complete MCP Tool & CLI Verb Reference

Generated by `buchcraft tools --markdown`. Do not edit by hand: a test fails when this file is out of date. Every tool is also a CLI verb (`buchcraft --help`). Dotted names (`app.verb`) are accepted.

Tool	What it does	Required	Optional
`buch_status`	Ledger status: counts, open drafts, unmatched bank lines, closed periods, demo flag.	-	-
`buch_accounts_list`	List the chart of accounts (EKR).	-	-
`buch_accounts_seed`	Insert the EKR seed chart (existing numbers untouched).	-	-
`buch_account_add`	Add an account.	`number`, `name`, `kind`	`tax_key`
`buch_entry_book`	Book a balanced journal entry. Net lines carry a tax_key; the engine generates the tax lines. Lines (incl. generated tax) must sum to zero.	`date`, `kreis`, `description`, `lines`	`beleg_nr`
`buch_entry_book_gross`	Book a gross amount in one step: account gets net, engine adds tax, contra gets -gross.	`date`, `kreis`, `description`, `account`, `contra`	`beleg_nr`, `gross`, `gross_cents`, `tax_key`
`buch_entry_reverse`	Storno an entry (the only way to correct a booking).	`id`, `date`, `reason`	-
`buch_entry_show`	Show one entry with its lines.	`id`	-
`buch_journal`	Journal for a date range, optionally one Belegkreis.	`from`, `to`	`kreis`, `limit`
`buch_balance`	Saldenliste (debit, credit, balance per account) for a date range.	`from`, `to`	-
`buch_account_ledger`	Kontoblatt with opening and running balance.	`account`, `from`, `to`	-
`buch_u30`	U30 VAT pre-fill (Kennzahlen 000/006/022/029/060/065/070/072/095) for a month YYYY-MM. Not a filing.	`month`	-
`buch_period_close`	Close a month (YYYY-MM); later bookings dated in it are refused.	`month`	-
`buch_bank_import`	Import a bank statement CSV (German or English headers). Re-importing the same file adds nothing. Needs --allow-path.	`path`	-
`buch_bank_unmatched`	Bank lines not yet booked.	-	`limit`

Tool	What it does	Required	Optional
`buch_bank_book`	Book an unmatched bank line against an account (gross; tax key splits it) and link it.	`id`, `account`	`description`, `tax_key`
`buch_drafts`	Drafts created from other apps' events (open only unless all=true).	-	`all`
`buch_draft_approve`	Book a draft.	`id`	-
`buch_draft_discard`	Discard a draft.	`id`	-
`buch_inbox_drain`	Read events from other craft apps (e.g. assetcraft depreciation) and turn them into drafts.	-	-
`buch_export_csv`	Export journal lines as semicolon CSV (DE format). Needs --allow-path.	`from`, `to`, `path`	-
`buch_demo_seed`	Fill an EMPTY ledger with a labelled demo company (October 2026).	-	-